Reactivity impacts of xenon and samarium

NE 630 | Nuclear Reactor Theory

22 LESSON

Read: FNRP § 10.1–10.3; Eqs. (10.4), (10.7), and (10.12)–(10.13).

Objective: Quantify the impact of xenon and samarium buildup on reactivity.

1 Review: reactivity

The multiplication factor (k) and reactivity (ρ) are related by

$$\rho = \frac{k-1}{k} \qquad \text{and} \qquad k = \frac{1}{1-\rho}.$$

Reactivity is given in $\Delta k/k$, $\%\,\Delta k/k$, pcm, or dollars. Changes are characterized by **(integral) worths**, $\Delta\rho$, and **reactivity coefficients**,

$$\alpha_x = \left.\frac{\partial \rho}{\partial x}\right|_{\text{non-}x\text{ fixed}}.$$

2 Reactivity in time

As a reactor operates at power, its excess reactivity typically *decreases* because ^{235}U is consumed and parasitic fission products (FPs) are born. That excess reactivity must be balanced by control.

To represent these processes, define a time-varying multiplication factor as

$$k(t) = \frac{\nu\bar{\Sigma}_{fT}^{F}(t)\; p\, \varepsilon\, P_{\text{NL}}}{\bar{\Sigma}_{aT}^{F}(t) + \bar{\Sigma}_{aT}^{\text{FP}}(t) + \varsigma\frac{V_M}{V_F}\left[\bar{\Sigma}_{aT}^{M} + \bar{\Sigma}_{aT}^{\text{control}}(t)\right]}, \qquad \text{(FNRP 10.4*)}$$

where P_{NL} is the probability that neutrons do not leak from the system.

During operation, $\bar{\Sigma}_{aT}^{\text{control}}(t)$ must be adjusted to maintain $k(t) = 1$. This model places adjustable control in the moderator volume, consistent with typical LWRs, whether by way of changing rod position or the soluble poison concentration.

3 Fission products and fission yields

Fission of ^{235}U (or ^{239}Pu, …) produces two massive fragments called **fission products**. Often, FPs are

- *radioactive* (which decay modes dominate?);
- *parasitic neutron absorbers* with large σ_a (thermal and/or epithermal).

The **fission yield**, γ_{fp}, specifies the production of a given FP per fission.

Nuclide	γ_{fp}	Nuclide	γ_{fp}
^{134}Xe	0.078	^{137}Cs	0.063
^{133}Cs	0.067	^{135}I	0.063
^{95}Mo	0.065	^{99}Tc	0.061

Selected ^{235}U yields from the IAEA WIMS Library Update project.

Ponderable: $\sum_{\text{fp}\,\in\,\text{all FPs}} \gamma_{\text{fp}} = ?$

4 Evolution of FP concentrations

For a FP produced directly from fission, production competes with radioactive decay and neutron absorption:

$$\frac{dN_{\text{fp}}}{dt} = -\lambda_{\text{fp}} N_{\text{fp}}(t) - \overbrace{\bar{\sigma}_a^{\text{fp}} N_{\text{fp}}(t)\,\phi}^{\bar{\Sigma}_a^{\text{fp}}(t)} + \gamma_{\text{fp}}\bar{\Sigma}_f\phi, \qquad \text{(FNRP 10.7)}$$

where, as usual, the $\bar{\ }$ means spectrum averaging over *all energies*, i.e.,

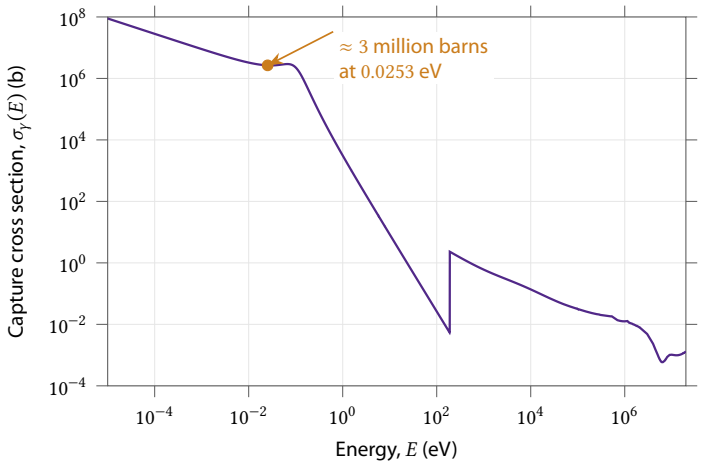
$$\bar{\sigma}_a^{\text{fp}} = \frac{\int_0^\infty \sigma_a^{\text{fp}}(E)\phi(E)\,dE}{\int_0^\infty \phi(E)\,dE}.$$

N_{fp} is a number density (cm^{-3}), λ_{fp} is a decay constant (s^{-1}), and the integrated flux ϕ has units cm^{-2}s^{-1}. Each term in the balance is a production or loss rate per unit volume.

5 The most important duo: I and Xe

Of all FPs, ^{135}Xe is especially important because of its enormous low-energy absorption cross section, but its direct yield is only $\gamma_{Xe} \simeq 0.00258$.

However, ^{135}Xe is also a daughter of ^{135}I, whose yield is $\gamma_I \simeq 0.063$.



Xenon-135 capture cross section (ENDF/B-VIII.0, 294 K; MT=102).

Neglecting absorption in ^{135}I, the evolution equations are

$$\frac{dN_I}{dt} = -\lambda_I N_I(t) + \gamma_I\bar{\Sigma}_f\phi, \qquad \text{(FNRP 10.12)}$$

$$\frac{dN_{Xe}}{dt} = -\lambda_{Xe}N_{Xe}(t) + \lambda_I N_I(t) + \gamma_{Xe}\bar{\Sigma}_f\phi - \bar{\sigma}_a^{Xe}N_{Xe}(t)\phi. \qquad \text{(FNRP 10.12)}$$

6 Equilibrium xenon worth

For constant $\bar{\Sigma}_f$ and ϕ, the concentrations reach saturation levels with zero time derivatives:

$$\begin{aligned} 0 &= -\lambda_I N_I(\infty) + \gamma_I \bar{\Sigma}_f \phi \\ 0 &= -\lambda_{Xe} N_{Xe}(\infty) + \lambda_I N_I(\infty) \\ &\quad + \gamma_{Xe} \bar{\Sigma}_f \phi - \bar{\sigma}_a^{Xe} N_{Xe}(\infty) \phi. \end{aligned}$$

Thus,

$$N_I(\infty) = \frac{\gamma_I \bar{\Sigma}_f \phi}{\lambda_I} \quad \text{and} \quad N_{Xe}(\infty) = \frac{(\gamma_I + \gamma_{Xe}) \bar{\Sigma}_f \phi}{\lambda_{Xe} + \bar{\sigma}_a^{Xe} \phi}.$$

Example: Suppose we have $\phi = 2 \times 10^{14}$ cm^{-2}s^{-1}, $\bar{\Sigma}_f = 0.16$ cm^{-1}, and $\bar{\sigma}_a^{Xe} = 2.6 \times 10^6$ b. Compute $\bar{\Sigma}_a^{Xe}$.

What is the equilibrium Xe worth for an initially critical reactor with $\bar{\Sigma}_a = 0.34$ cm^{-1}?

7 The xenon transient: an initial-value problem

The solution of FNRP Eqs. (10.12) and (10.13) is straightforward for a constant flux using normal techniques, but when the flux varies in time, numerical techniques are useful. Almost all tools for solving systems of initial-value problems require that we write the problem in the form

$$\dot{\mathbf{y}} = \mathbf{f}(t, \mathbf{y}(t)), \quad \mathbf{y}(0) = \mathbf{y}_0 .$$

For our problem, that looks like

$$\overbrace{\frac{d}{dt}\begin{pmatrix} N_I \\ N_{Xe} \end{pmatrix}}^{\mathbf{y}(t)} = \overbrace{\begin{pmatrix} -\lambda_I & 0 \\ \lambda_I & -\lambda_{Xe} - \bar{\sigma}_a^{Xe}\phi(t) \end{pmatrix}\begin{pmatrix} N_I \\ N_{Xe} \end{pmatrix} + \begin{pmatrix} \gamma_I \bar{\Sigma}_f \phi \\ \gamma_{Xe} \bar{\Sigma}_f \phi \end{pmatrix}}^{\mathbf{f}(t,\mathbf{y}(t))}$$

with initial concentration $N_{I,0}$ and $N_{Xe,0}$.

```
 import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

def dNdt(t, N, phi, sig_a_Xe, Sigma_f):
  gamma_I, gamma_Xe = 0.0639, 0.00237
  lambda_I, lambda_Xe = 0.693/(6.7*3600),
    0.693/(9.2*3600)
  sig_a_Xe = sig_a_Xe*1e-24
  N_I, N_Xe = N
  dN_Idt = -lambda_I*N_I + gamma_I*Sigma_f*phi(t)
  dN_Xdt = -lambda_Xe*N_Xe + lambda_I*N_I + \
           (gamma_Xe*Sigma_f-sig_a_Xe*N_Xe)*phi(t)
  return dN_Idt, dN_Xdt

hour=3600; day = 24*hour;

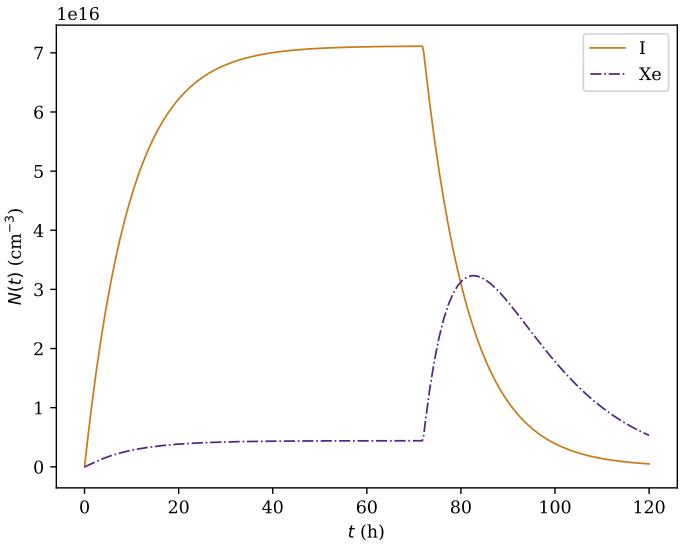
flux = lambda t: 2e14 if t < 3*day else 0.0

result = solve_ivp(fun=dNdt, t_span=(0.0, 5*day),
  y0=(0, 0), args=(flux, 2.3e6, 0.16), max_step=1*hour)
t = result.t/3600; N_I, N_Xe = result.y

orange, purple = "#CA7C1B", "#512888"
plt.plot(t, N_I, color=orange, lw=1, label="I")
plt.plot(t, N_Xe, color=purple, lw=1, ls="-.", label="Xe
    ")
plt.xlabel("$t$ (h)"); plt.ylabel("N(t) (cm$^{-3}$)")
plt.legend()
plt.show()
```

Numerical solution of the xenon transient.

8 The result



Results of the xenon transient simulation. Shutdown occurs at 72 h.

Why does xenon rise after shutdown?
Setting $\phi = 0$ removes direct fission production and neutron absorption. The balances become

$$\begin{aligned} \frac{dN_I}{dt} &= -\lambda_I N_I, \\ \frac{dN_{Xe}}{dt} &= -\lambda_{Xe} N_{Xe} + \lambda_I N_I. \end{aligned}$$

Iodine continues to feed xenon after the neutron-absorption sink disappears. Xenon therefore initially rises in this experiment, peaks, and then decays as the iodine reservoir is exhausted.

Ponderable: At the xenon peak after shutdown, what relation holds between $\lambda_I N_I$ and $\lambda_{Xe} N_{Xe}$? How does the changing xenon inventory alter the control needed for restart?

9 Beyond Xe!

Here, we've emphasized ^{135}Xe due to its strong impact on reactivity and dynamics. Other strongly absorbing fission products are

- ^{149}Sm (see FNRP for a discussion), ^{147}Sm;
- ^{133}Cs;
- ^{147}Nd, ^{143}Nd;
- ^{103}Rh.

For full production/decay chains of important FPs, see the IAEA WIMS fission-product chains.